

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate CC-486/Onureg in Participants With Moderate or Severe Hepatic Impairment Compared With Normal Hepatic Function in Participants With Myeloid Malignancies **Short Title/Acronym:** Pharmacokinetics of CC-486 in Hepatic Impairment **Protocol Number:** CA055-001 **NCT Number:** NCT05209295 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** CC-486 / Onureg / Vidaza (oral azacitidine) | ATC Code: L01BC07 **Phase of Development:** Phase 1 **Trial Status:** Active, not recruiting **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of moderate or severe liver impairment on the drug levels and pharmacokinetics of oral azacitidine (CC-486) in participants with myeloid malignancies. **Secondary Objectives:** To evaluate the safety and tolerability of oral azacitidine in participants with myeloid malignancies who have varying degrees of hepatic impairment.

2.2 Background & Rationale To address the gap in pharmacokinetic and safety data for CC-486 (oral azacitidine) in patients with myeloid malignancies who concurrently suffer from moderate to severe hepatic impairment. Because liver function can significantly impact the metabolism and toxicity of oncological therapies, this study aims to determine if moderate or severe liver issues influence how the body processes the drug to establish safe dosing guidelines for this vulnerable subpopulation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel Assignment (Normal Hepatic Function vs. Moderate/Severe Hepatic Impairment) **Blinding:** Open-label **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 32 participants **Number of Sites:** Multicenter **Estimated Study Start:** Early 2022 **Estimated Primary Completion:** Active, not recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Documented diagnosis of Myelodysplastic syndrome, Acute myeloid leukemia, Non-acute promyelocytic leukemia, Chronic myelomonocytic leukemia, Philadelphia-negative myeloproliferative neoplasms, Myelodysplastic syndrome Myeloproliferative neoplasms overlap, Accelerated phase and blast phase Myeloproliferative neoplasms, Blastic plasmacytoid dendritic cell neoplasm according to the World Health Organization (WHO) 2016 classification. 2. Life expectancy of $\geq$ 3 months. 3. Stable renal function without dialysis for at least 2 months prior to investigational product administration. 4. Has moderate or severe hepatic impairment as defined by National Cancer Institute Organ Dysfunction Working Group criteria.

4.2 Exclusion Criteria 1. Chemotherapy or radiotherapy within 2 weeks or 5 half-lives, whichever is longer, prior to the first day of investigational product administration. 2. Persistent, clinically significant non-hematologic toxicities from prior therapies which have not recovered to $<$ Grade 2. 3. Any condition including the presence of laboratory abnormalities, which places the participant at unacceptable risk if he/she were to participate in the study. 4. History of inflammatory bowel disease, celiac disease, prior gastrectomy, gastric bypass, upper bowel removal, or any other gastrointestinal disorder or defect that would interfere with the absorption of the investigational product and/or predispose the participant to an increased risk of gastrointestinal toxicity.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: CC-486 (Onureg) administered to defined parallel groups based on baseline liver function (Normal vs. Moderate Impairment vs. Severe Impairment). **Route of Administration:** Oral **Duration of Treatment:** Administered in standard treatment cycles until disease progression, unacceptable toxicity, or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for underlying myeloid malignancies. **Prohibited:** Concurrent investigational therapies and strong CYP inhibitors/inducers that could interfere with the pharmacokinetic profiling of oral azacitidine.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Hepatic/Renal Function Labs, Disease Staging
Baseline	Day 0	Group Assignment based on Hepatic Function,	

First Dose CC-486 | | **Interim** | Cycle 1 & Beyond | Intensive PK Blood Draws, Safety Labs, Efficacy/Tolerability Assessment | | **End of Study** | Follow-up | Final PK evaluation, Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Pharmacokinetic (PK) parameters of oral azacitidine (CC-486) evaluated via: 1. **AUC_{0-t}**: Estimation of area under the plasma concentration-time curve (AUC) calculated from time zero to the last measured time point. 2. **AUC_{0-∞}**: Estimation of AUC calculated from time zero to infinity. 3. **C_{max}**: Observed maximum concentration. **Measurement Method:** Serial blood sampling and central laboratory PK analysis.

7.2 Secondary Endpoints Definition: Overall safety and tolerability evaluated via: 1. Incidence of adverse events. 2. Incidence of serious adverse events. 3. Number of participants with clinically significant changes in electrocardiogram parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for hematologic toxicities (e.g., cytopenias), gastrointestinal events, and specifically hepatotoxicity via laboratory assessments and patient reporting. **Serious Adverse Events (SAEs):** Routine reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review by the sponsor to continually assess dose-limiting toxicities in the hepatic impairment cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set for primary drug level evaluation; Safety Set for all participants who received at least one dose of the investigational product. **Sample Size Justification:** Descriptive Phase 1 PK trial with a targeted enrollment of 32 participants; formal power calculations are typically bypassed in favor of clinical PK evaluation standards. **Handling of Missing Data:** Standard non-compartmental analysis (NCA) imputation methods for missing PK timepoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict onsite monitoring for PK sample

integrity, timing precision, and AE/SAE verification. **Audit Trail:** Standard eCRF implementation with data clarification mechanisms for central PK labs.

11. Study Identifiers & Governance

Primary ID: CA055-001 **Registry Number:** NCT05209295 **Posting ID:** 1147817936519647759 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate CC-486/Onureg in Participants With Moderate or Severe Hepatic Impairment Compared With Normal Hepatic Function in Participants With Myeloid Malignancies **Official Regulatory Title:** A Phase 1, Multicenter, Open-label Study to Evaluate the Pharmacokinetics of CC-486 (Onureg®) in Subjects With Moderate or Severe Hepatic Impairment Compared With Normal Hepatic Function in Adult Subjects With Myeloid Malignancies **Disease Areas / Conditions:** Hepatic Insufficiency, Neoplasms

12. Clinical Framework (Myeloid Malignancies & Hepatic Impairment Specific)

Primary Conditions: Neoplasms (Myeloid Malignancies) and Hepatic Insufficiency. **Diagnosis Threshold:** WHO 2016 classification standards; moderate/severe hepatic impairment defined by National Cancer Institute Organ Dysfunction Working Group criteria. **Medical Methodology:** Pharmacokinetic profiling of hypomethylating agent therapy (oral azacitidine). **Optimization Target:** Establishing safe dosage parameters in patients with compromised liver clearance.

13. Study Procedures & Methodology

Management Pathway: Specialized Phase 1 PK sampling and safety monitoring pathway. **Education Protocol (HCP):** Investigator training on rigid serial PK blood draw timing and handling. **Education Protocol (Patient):** Informed consent detailing the rigorous blood sampling schedule and reporting of potential hepatic or hematologic adverse events. **Quality Audit Mechanism:** Close PK sample integrity tracking and adherence to liver function stratification guidelines.

14. Observation & Follow-Up Schedule

Total Duration: Active, not currently recruiting. **Onsite Visit Cadence:** Intensive monitoring during Cycle 1 for PK parameter mapping. **Remote Monitoring Frequency:** Inter-cycle safety checks as permitted by Phase 1 guidelines. **Checklist Review Interval:** At the beginning and end of each standard treatment cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					